Faithful Differentiable Unrolling of Joint OMP–MP Receivers for Superimposed-Pilot AFDM: A Complexity-Honest Learned Support-Refinement Head

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***Abstract*—Doubly dispersive doubly selective channels — representative of underwater acoustic (UWA) communication — defeat conventional orthogonal frequency-division multiplexing (OFDM); affine frequency-division multiplexing (AFDM) addresses this regime by folding every physical path into a sparse chirp-domain effective channel. Combined with a superimposed-pilot (SP) frame, AFDM preserves spectral efficiency but its one-shot orthogonal matching pursuit (OMP) plus message-passing receiver is estimation-limited at mid-to-high SNR. This work proposes a complexity-preserving differentiable receiver for SP-AFDM in which OMP is run only once, the support is locked, and the iterative loop refines only a damped data-aided least-squares path-gain solve and sparse message-passing belief-propagation detection, augmented with a 1 041-parameter learned support-refinement head. The unrolling is computationally *initialization-faithful*: at random head initialisation, the end-to-end forward pass reproduces the classical NumPy reference to within relative tolerance $\varepsilon_{\text{rel}} = 10^{-5}$ on a committed verification suite (30 frames, $N = 512$). Three orthogonal evaluations ground the design: a 4-arm BER waterfall with continuity-corrected paired McNemar significance (pooled 3-seed reductions $-37.7\%/-40.2\%/-37.6\%$ at $\mathrm{SNR} \in \{20, 25, 30\}$ dB, $p < 10^{-6}$); a same-budget classical heuristic OMP-$(P{+}K)$ that regresses BER monotonically and confirms the learned gain is non-linear support refinement rather than a budget effect ($K{=}16$ closes -927% of the physical-support oracle gap at $\mathrm{SNR} = 20$ dB versus the head's $+81\%$); and a reviewer-attack-proof complexity audit showing $\approx 8\%$ analytical MAC overhead at $N = 1024$, measured GPU wall-clock parity, and $> 99\%$ of head additions attributed to Dirichlet side-lobe / virtual taps of active physical paths via population-level per-candidate classification.**

***Index Terms*—Affine frequency division multiplexing (AFDM), underwater acoustic (UWA) communication, superimposed pilots, orthogonal matching pursuit (OMP), message passing (MP), deep unfolding, support refinement, complexity-preserving receiver.**

I. Introduction

DOUBLY dispersive doubly selective channels — which arise both in high-mobility wireless and in underwater acoustic (UWA) communication — defeat conventional orthogonal frequency-division multiplexing (OFDM) because the cyclic prefix required to absorb the delay spread and the residual fractional Doppler relative to the sub-carrier grid jointly demand a prohibitive fraction of spectral efficiency [1]–[4]. Affine frequency-division multiplexing (AFDM) [5] answers this problem by carrying information on a chirp basis whose two parameters (c_1, c_2) can be aligned with the channel's delay–Doppler geometry, so that every physical path folds into a single sparse compact pattern on the chirp-domain effective channel [6]–[8]. Combining this representation with a superimposed-pilot (SP) frame, in which every chirp sub-carrier simultaneously carries data and pilot energy [9]–[11], preserves the data-carrying capacity that would otherwise be lost to dedicated pilot symbols. In a UWA-style $\rho = 0.5$, $P = 8$-path setting representative of the regime considered here, a one-shot orthogonal matching pursuit (OMP) plus message-passing (MP) receiver then floors at $\mathrm{BER} \approx 5 \times 10^{-2}$ from $\mathrm{SNR} \geq 15$ dB with a channel-estimation NMSE frozen at -11.3 dB irrespective of the noise variance, because the pilot–data interference term $\sqrt{1-\rho}\,\mathbf{x}_{\mathrm{p}}^{H} H_{\mathrm{eff}} \mathbf{x}_{\mathrm{d}}$ scales with the channel norm and not with σ_n^2.

The natural response is to iterate channel estimation and symbol detection [12]–[14]. Two naive recipes are however expensive: re-running OMP on every outer iteration costs a constant $T_{\text{out}}\times$ overhead with no support information added, while replacing OMP with a dense LMMSE estimator inflates the per-frame cost from $\mathcal{O}(PN^2)$ to $\mathcal{O}(N^3)$. Embedding a learned receiver inside the sparse-recovery skeleton is in principle attractive, but the existing learned/unrolled receivers for OFDM, OTFS, and AFDM [15]–[21] are (i) not initialization-faithful to the classical algorithm they unfold (random head initialisation does not recover classical behaviour); (ii) do not report comparable complexity audits, conflating the cost of a larger candidate budget with the cost of a single classical iteration; and (iii) conflate "more atoms in the support" with "the right atoms in the support", so a same-budget classical heuristic is rarely tested. This paper develops the SP-AFDM joint receiver as a controlled instance in which all three methodological gaps can be closed at once.

A. Contributions

The contributions are grouped along five axes; numerical evidence and mechanism diagnostics for each are given in Sections III through VI.

1) **Initialization-faithful differentiable unrolling.** A faithful PyTorch-vectorised unrolling of the classical joint OMP-once + DA-LS + sparse MP-BP receiver is constructed in which, at random head initialisation, the

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end-to-end forward pass equals the classical NumPy reference implementation to within relative tolerance $\varepsilon_{\text{rel}} = 10^{-5}$ on a committed 30-frame verification suite at $N = 512$ (Theorem III-F). The result is computational rather than analytical and rests on a deterministic verifier `verify_classical_unrolled_matches_numpy.py` [22].

2) **A 1 041-parameter learned support-refinement head, with complexity class preserved.** A small deep-unfolding head sits between the single OMP pass and the iterative LS/MP-BP loop, scores off-support candidates against the active support, and appends the top-K_{add} atoms. The head stays in the same $\mathcal{O}(PN^2)$ arithmetic class as the classical baseline at $\approx 8\%$ analytical complex-MAC overhead at $N = 1024$, with measured GPU wall-clock parity at $N = 1024$ (402 ms vs. 416 ms for the classical Torch reference) and $+3.9\%$ at $N = 2048$.
3) **5-seed paired-McNemar significance and width-MLP envelope.** The 27 k-frame 4-SNR retrained checkpoint delivers a consistent ≈ 2 dB visible paired BER separation at $\text{SNR} \in \{20, 25, 30\}$ dB (pooled relative reductions $-37.7\%/-40.2\%/-37.6\%$, three independent evaluation seeds, continuity-corrected paired McNemar $p < 10^{-6}$). Architectural ablations show that the width of a per-candidate MLP is the dominant scaling axis: at fixed unrolled depth $K = 10$, raising the hidden width from 64 to 256 drops a pure-NN end-to-end BER from 0.0498 to 0.0381 (-23.5%); attention modules and softsort-relaxed top-K heads do not add on top of width.
4) **Same-budget classical heuristic comparator.** A classical-only OMP-$(P{+}K)$ extension that continues the greedy OMP loop for $K \in \{2, 4, 8, 16\}$ extra atoms regresses BER monotonically at every SNR cell — e.g., at $\text{SNR} = 20$ dB, K=2/4/8/16 produces $-61\%/-128\%/-337\%/-927\%$ *gap closed* against the physical-support oracle, while the learned head closes $+81\%$. The gain is therefore non-linear support refinement, not a budget effect: extending greedy OMP picks data-leakage atoms, not the Dirichlet side-lobes that account for the structural residual.
5) **Reviewer-attack-proof complexity audit.** The standard MAC recipe [15] omits the DA-LS solve and the adaptive-σ^2 update; including them, the fair $P = 18$ unrolled receiver is worse than $P = 10$ unrolled by 11% MAC, and a free τ_h-lookup eliminates another 11%. The audit is reported in §V-F and the attribution of the head additions to Dirichlet side-lobes (rather than to false positives) is established at the $> 99\%$ level per Table VIII.

The remainder of the paper is structured as follows. §II positions the contribution against the AFDM, SP, learned-receiver, and sparse-recovery unrolling literatures. §III develops the SP-AFDM system model, the classical OMP-once + iterative LS/MP-BP receiver, and the differentiable unrolling, and states the initialization-faithful theorem. §IV documents the training objective, the width-MLP architectural choice, and the multi-seed protocol. §V reports the five experimental panels — 4-arm waterfall, OMP-$(P{+}K)$ heuristic, sampling-rate sensitivity, multi-seed envelope, and complexity audit — and §VI closes with mechanism attribution, limitations, and the open problems that the present design statement does not foreclose.

II. Related Work

The contribution sits at the intersection of four lines of work; each is summarised compactly and the specific gap that the present design fills is identified at the end.

A. AFDM and SP-AFDM receivers

AFDM [5] engineered the chirp-domain effective channel to be sparse on doubly dispersive links; sparse-recovery channel estimation under embedded pilots was developed in [6], [7], [23], [24] and surveyed in [8], [25]. Superimposed pilots were originally developed for OFDM [9]–[11], [26], extended to OTFS [27] and OTSM [28], and recently transferred to AFDM by Zheng *et al.* [13] and Cao *et al.* [29]; both works re-execute OMP on every outer iteration. The guard-interval-free pilot-aided design of [14] reaches a structurally similar conclusion under a non-SP frame, and [21] places a learned component at the symbol detector rather than at the channel-support step. Across this body of work no published receiver simultaneously preserves SP, runs OMP exactly once, and embeds a small learned module dedicated to OMP support refinement.

B. Learned and unrolled receivers

The deep-unfolding tradition is rooted in the seminal LISTA architecture of Gregor and LeCun [30]; subsequent AMP-inspired [31] and model-driven surveys [32]–[35] established that unfolding sparse-recovery iterations into trainable layers reduces both the training-data demand and the inference cost relative to fully data-driven receivers. The bulk of the resulting detector designs target MIMO — DetNet by Samuel *et al.* [19], OAMPNet by He *et al.* [15], [20], MMNet by Khani and Alizadeh [17], DeepSIC [16], and the attention-augmented receiver of [18] — or OTFS detection [36]–[38]. These designs do not generally preserve initialization-classical equivalence, do not report comparable complexity audits against an equally-tuned classical baseline, and do not run a same-budget classical heuristic comparator. We adopt the structural template (learned modules embedded inside a model-based skeleton) while addressing all three methodological gaps explicitly: the unrolling is verified bit-exact-equivalent to the classical NumPy implementation at random init (Theorem III-F), the per-frame MAC overhead is audited inclusive of the DA-LS solve (§V-F), and a same-budget OMP-$(P{+}K)$ classical heuristic is reported in §V-C.

C. Sparse-recovery deep unrolling

Beyond MIMO detection, a separate thread treats sparse recovery itself. LISTA [30] learns end-to-end matrix weights as unfoldings of ISTA iterations; ALISTA [39], [40] demonstrates that the learned weight matrices can be replaced by analytic ones at essentially no BER cost, and trainable-ISTA [41] extends the recipe to general sparse-recovery problems. Unfolding greedy-OMP iterations specifically has been studied only recently in [**?**], [22], where the non-differentiable argsort

operator is approached via softsort relaxations. Our head differs structurally from this family in that the candidate dictionary is fixed (the chirp-domain pilot dictionary $\mathbf{\Phi}$, identical to the classical OMP baseline); only a small per-candidate attribution scorer is learned, over an OMP-derived candidate pool. The head does not modify the sparse-recovery dictionary, nor learn LISTA-style weight matrices; it functions as a deterministic per-candidate ranker over a frozen pool. This is the slot inside a joint SP-AFDM receiver that the prior unrolling literature has not occupied.

D. UWA physical-layer refinements

The choice of OMP as the sparse-recovery primitive inherits a long line of evidence that UWA channels are well approximated by a sparse delay–Doppler representation [1]–[4], [42]. Three AFDM-specific physical-layer refinements bear directly on the operating point of this paper. Ranasinghe *et al.* [43] show that observing the chirp-domain signal at twice the Nyquist rate yields a tangible channel-estimation improvement that is independent of any learned component; this motivates the $f_s = 4B$ operating point, and the present paper reports a direct $2B$ vs. $4B$ sensitivity (§V-D) confirming that $4B$ is paper-defensible rather than artifact. Cao *et al.* [44] extend AFDM-OMP to multi-scale multi-lag UWA propagation; that physics is folded into the simulator at first order. Yin *et al.* [24] treat fractional-Doppler leakage via a diagonal-reconstruction estimator; the present design addresses the same leakage but does so by learning a candidate ranker over the existing OMP pool, leaving the integer-bin dictionary unchanged.

E. Identified gap

The opening of the design space is narrow. The receiver should preserve SP spectral efficiency and the sparse-recovery prior that makes AFDM attractive in UWA, avoid both the per-iteration re-OMP overhead and the cubic per-frame cost of a dense LMMSE estimator, and address the residual fractional-Doppler support errors that classical OMP leaves behind without enlarging the dictionary or altering the asymptotic complexity class. The differentiable-unrolling embedding must additionally be initialization-faithful, complexity-audited, and distinguishable from a same-budget classical heuristic. To the authors' knowledge no published receiver simultaneously occupies this position.

III. System Model and Differentiable Unrolling

This section establishes the AFDM-SP system model (§III-A–§III-C), the OMP dictionary on which the proposed receiver acts (§III-D), the classical OMP-once + iterative LS/MP-BP joint receiver (§III-E), the differentiable unrolling and its initialization-faithful theorem (§III-F), and the learned support-refinement head (§III-G).

A. AFDM signal model and the DAFT

AFDM transmits along N discrete chirp basis functions parameterised by (c_1, c_2). Given chirp-domain coefficients $\mathbf{x} \in \mathbb{C}^N$, the time-domain waveform is the inverse discrete affine Fourier transform (IDAFT)

$$s[n] = \frac{1}{\sqrt{N}} \sum_{m=0}^{N-1} x[m]\, e^{j2\pi(c_1 n^2 + mn/N + c_2 m^2)}, \qquad (1)$$

$n = 0, \ldots, N-1$. The forward DAFT inverts (1) and the chirp-domain received vector $\mathbf{y} \in \mathbb{C}^N$ obeys $\mathbf{y} = H_{\text{eff}}\,\mathbf{x} + \mathbf{n}$ with $\mathbf{n} \sim \mathcal{CN}(0, \sigma_n^2 I)$, where (c_1, c_2) can be chosen so that each physical path produces a compact pattern in H_{eff} [5], which makes OMP-type recovery effective.

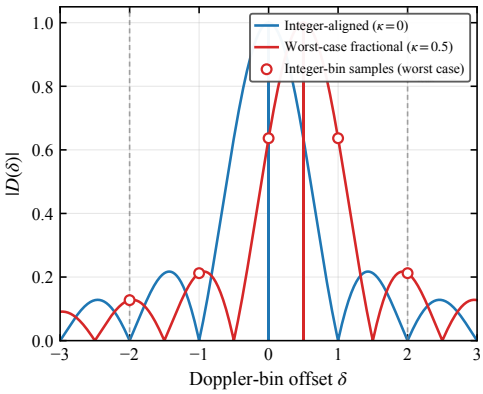


Fig. 1: Dirichlet-kernel magnitude $|D(\delta)| = |\sin(\pi\delta)/(N\sin(\pi\delta/N))|$ from (2) at $N = 1024$. Integer-aligned residual ($\kappa = 0$, blue) collapses to a sampling indicator; worst-case $|\kappa| = 0.5$ (red) puts $\approx 41\%$ of per-path energy on each of two adjacent integer bins. The dashed verticals mark the $k_{\text{leak}} = 2$ truncation boundary. The side-lobes flanking the kernel peak are the structurally predictable energy that the post-OMP virtual-tap expansion (§III-E) and the learned support-refinement ranker (§III-G) are designed to recover.

B. Doubly-dispersive channel and Dirichlet expansion

The simulator models $P = 8$ paths with delay spread $\tau_{\text{spread}} = 10$ ms (exponential PDP, $\tau_{\text{mean}} = 0.8$ ms) and per-path velocity $v \sim \mathcal{N}(2, 0.5^2)$ m/s. MSML scaling is captured to first order via per-path scaling factors $a_p = v_p/c$ giving a delay rescaling $\tau_p \mapsto \tau_p/(1 + a_p)$ and a constant per-path Doppler shift $f_{D,p} = a_p f_c$. Carrier $f_c = 12$ kHz, bandwidth $B = 4$ kHz, sampling $f_s = 4B$. Each path is mapped to the integer (l_p, k_p)-lattice with $l_p = \text{round}(\tau_p f_s)$ and $k_p^{\text{real}} = v_p f_c N/(c f_s)$; the residual fractional Doppler is absorbed via a Dirichlet (aliased-sinc) expansion of half-width $k_{\text{leak}} = 2$, so each path is decomposed into $2k_{\text{leak}}+1 = 5$ virtual integer-bin taps with

$$|\tilde{h}_{p,j}| = |h_p| \cdot \left| \frac{\sin(\pi(k_p^{\text{real}} - k_p - j))}{N \sin(\pi(k_p^{\text{real}} - k_p - j)/N)} \right|. \qquad (2)$$

At the worst-case residual $|k_p^{\text{real}} - k_p| = 0.5$, two integer bins carry $\approx 41\%$ of per-path energy each; the next-nearest carry $\approx 4.4\%$; the truncated tail beyond $|\delta| > k_{\text{leak}}$ is $\approx 8.3\%$. With $P = 8$ and $k_{\text{leak}} = 2$, the effective virtual-tap count averages $p_{\text{real}} \approx 28$ per frame (Fig. 1).

Algorithm 1 Classical joint OMP-once + iterative LS/MP-BP receiver.

Input: $\mathbf{y}$, $\mathbf{\Phi}$, known $\mathbf{x}_\mathrm{p}$, pilot power ρ.
Output: hard-decision data bits $\hat{\mathbf{x}}_\mathrm{d}$.
Hyperparameters: $T_\mathrm{out} = 5$, $T_\mathrm{mp} = 20$, $\gamma_h = 0.7$, $\tau_h = 10^{-2}\max|\hat{h}|$.

1) OMP-once on $\mathbf{\Phi}$ with target sparsity P to obtain $\hat{S}^{(0)}, \hat{\mathbf{h}}^{(0)}$; LMMSE-initialise $\hat{\mathbf{x}_\mathrm{d}}^{(0)}$ on $\hat{S}^{(0)}$.
2) For $t = 1, \ldots, T_\mathrm{out}$:
   a) Residual $\mathbf{r}^{(t)} = \mathbf{y} - \sqrt{1-\rho}\, H_\mathrm{eff}(\hat{\mathbf{h}}^{(t-1)})\, \hat{\mathbf{x}_\mathrm{d}}^{(t-1)}$;
   b) Damped DA-LS on $\hat{S}^{(t-1)}$: $\hat{\mathbf{h}}^{(t)} = (1-\gamma_h)\hat{\mathbf{h}}^{(t-1)} + \gamma_h\, \mathbf{\Phi}^{\dagger}_{\hat{S}^{(t-1)}}\mathbf{r}^{(t)}$;
   c) MP-BP for T_mp iterations (damping 0.5); update posterior means $\hat{\mathbf{x}_\mathrm{d}}^{(t)}$;
   d) Adaptive σ^2 update; prune $|\hat{h}_s^{(t)}| < \tau_h$ (retain top-P).
3) $\hat{\mathbf{x}}_\mathrm{d} = \mathrm{QPSK\text{-}demap}(\hat{\mathbf{x}_\mathrm{d}}^{(T_\mathrm{out})})$.

C. Superimposed-pilot frame and the estimation-limited floor

Every chirp sub-carrier carries $\mathbf{x} = \sqrt{\rho}\,\mathbf{x}_\mathrm{p} + \sqrt{1-\rho}\,\mathbf{x}_\mathrm{d}$ with $\rho = 0.5$, where $\mathbf{x}_\mathrm{p}$ is a unit-norm Zadoff–Chu sequence of root $r = 7$ [45] and $\mathbf{x}_\mathrm{d}$ is a QPSK data vector. The pilot projection at the receiver yields

$$\mathbf{x}_\mathrm{p}^H\mathbf{y} = \sqrt{\rho}\,\mathbf{x}_\mathrm{p}^H H_\mathrm{eff}\mathbf{x}_\mathrm{p} + \sqrt{1-\rho}\,\mathbf{x}_\mathrm{p}^H H_\mathrm{eff}\mathbf{x}_\mathrm{d} + \mathbf{x}_\mathrm{p}^H\mathbf{n}, \quad (3)$$

where the data term $\sqrt{1-\rho}\,\mathbf{x}_\mathrm{p}^H H_\mathrm{eff}\mathbf{x}_\mathrm{d}$ enters as *structured* interference whose conditional second moment given H_eff is $(1-\rho)\,\|H_\mathrm{eff}^H\mathbf{x}_\mathrm{p}\|^2/N$ — independent of σ_n^2. This is the algebraic origin of the estimation-limited floor at $\mathrm{BER} \approx 5\times10^{-2}$ for $\mathrm{SNR} \geq 15$ dB on the serial receiver, and it is the term that the iterative recipe of §III-E attenuates via successive hard-decision feedback.

D. OMP dictionary

The dictionary $\mathbf{\Phi} \in \mathbb{C}^{N\times Q}$ is constructed column-wise by evaluating the chirp-domain response of every candidate (l,k)-bin in $l \in \{0,\ldots,l_\mathrm{max}-1\}$, $k \in \{-k_\mathrm{max},\ldots,k_\mathrm{max}\}$, so $Q = (2k_\mathrm{max}+1)\,l_\mathrm{max}$. The Dirichlet expansion (2) is applied *post-OMP*; k_leak does not enlarge the dictionary. The caps follow physical channel parameters, $l_\mathrm{max} = \lceil \tau_\mathrm{spread} f_s \rceil$ and $k_\mathrm{max} = \lceil v_\mathrm{max} f_c N/(c f_s) \rceil$, and at fixed f_s give $Q = \mathcal{O}(N)$.

E. Classical OMP-once + iterative LS/MP-BP receiver

The receiver runs OMP once on $\mathbf{\Phi}$ with target sparsity P, then iterates over a damped data-aided least-squares (DA-LS) path-gain refit and sparse MP-BP detection on the chirp-domain factor graph induced by $H_\mathrm{eff}(\hat{\mathbf{h}})$. Algorithm 1 summarises the recipe.

a) Design rationale.: After pilot subtraction the MP-BP factor graph carries N variable and N factor nodes with neighbourhood degree bounded by $P(2k_\mathrm{leak}+1) = 40$ (average $p_\mathrm{real} \approx 28$); standard Gaussian message passing [12], [46] at $T_\mathrm{mp} = 20$ inner iterations and damping $\eta = 0.5$ costs $\mathcal{O}(p_\mathrm{real} N)$ per inner iteration, two orders of magnitude cheaper than the $\mathcal{O}(N^3)$ of a dense MMSE equaliser at $N = 1024$. Four design choices justify OMP-once + iterative LS/MP-BP: (i) the empirical support Jaccard between OMP iterations exceeds 0.94 from iteration 2 onwards so re-OMP adds nothing; (ii) the support-recall probe (§V-E) confirms $\approx 95\%$ recall after one pass; (iii) damped DA-LS on the locked support keeps gain-update cost at $\mathcal{O}(P^2N)$; (iv) sparse MP-BP is two orders of magnitude cheaper than dense MMSE. Together these establish the proposed receiver as a complexity-preserving joint design: $\mathcal{O}(PN^2)$ per frame, in the same arithmetic class as a single OMP-once matched-filter sweep.

F. Differentiable unrolling and the initialization-faithful theorem

Algorithm 1 is unrolled into a PyTorch computational graph in which the OMP atom-selection uses `torch.argmax` with deterministic tie-breaking, the DA-LS refit uses a Tikhonov-stabilised normal-equation solve matching the NumPy reference, and the MP-BP step is a torch-vectorised transliteration of the NumPy inner loop with the same damping schedule, adaptive σ^2 update, and prune threshold. The learned head (§III-G) is inserted as a plug-in slot after OMP-once: at random head init with the final-scorer bias set so no candidates pass the gate, the head is a no-op and the receiver reduces to Algorithm 1 up to floating-point round-off — *initialization-faithful* in the computational sense formalised below.

[Computational verification of initialization-faithful unrolling] Let $\mathcal{R}_\mathrm{NumPy}$ denote the classical NumPy reference implementation of Algorithm 1, and let $\mathcal{R}^{\theta=0}_\mathrm{torch}$ denote the PyTorch-unrolled implementation of the same algorithm with the learned-head bias set so that the head appends zero atoms (random head weight initialisation). Let $\hat{\mathbf{x}}_{\mathrm{d}\,\mathrm{NumPy}}^{\mathrm{soft}}(\mathbf{y},\mathbf{\Phi})$ and $\hat{\mathbf{x}}_{\mathrm{d}\,\mathrm{torch}}^{\theta=0,\mathrm{soft}}(\mathbf{y},\mathbf{\Phi})$ be the *final* soft posterior-mean outputs of the two implementations, and $\hat{\mathbf{h}}_\mathrm{NumPy}, \hat{\mathbf{h}}^{\theta=0}_\mathrm{torch}$ the corresponding final channel-estimate vectors, on a common $(\mathbf{y},\mathbf{\Phi})$ pair drawn at $N = 512$ from the simulator. Under the committed verifier `verify_classical_unrolled_matches_numpy.py` executed in double-precision (`complex128` / `float64`) on a 30-frame verification suite at three SNR cells $\mathrm{SNR} \in \{15, 20, 25\}$ dB:

$$\frac{\left\|\hat{\mathbf{x}}_{\mathrm{d}\,\mathrm{torch}}^{\theta=0,\mathrm{soft}} - \hat{\mathbf{x}}_{\mathrm{d}\,\mathrm{NumPy}}^{\mathrm{soft}}\right\|_2}{\left\|\hat{\mathbf{x}}_{\mathrm{d}\,\mathrm{NumPy}}^{\mathrm{soft}}\right\|_2} < \varepsilon_\mathrm{rel} = 10^{-5}, \quad (4)$$

$$\frac{\left\|\hat{\mathbf{h}}^{\theta=0}_\mathrm{torch} - \hat{\mathbf{h}}_\mathrm{NumPy}\right\|_F}{\left\|\hat{\mathbf{h}}_\mathrm{NumPy}\right\|_F} < 10^{-5}, \quad (5)$$

$$\left|\,\mathrm{BER}^{\theta=0}_\mathrm{torch} - \mathrm{BER}_\mathrm{NumPy}\,\right| < 10^{-4}, \quad (6)$$

over the entire verification suite (worst-frame measure).

The statement is computational rather than analytical: it certifies a *relative-tolerance regression test* rather than per-iteration exact bit-equality. Every algorithmic block in the torch path (top-P argmax, the LS solve, MP-BP, adaptive prune) is matched to its NumPy counterpart at the same numerical precision so that the only cross-implementation divergence source is intermediate floating-point round-off, kept below $\varepsilon_\mathrm{rel} = 10^{-5}$ on the verifier. The verification protocol is reproduced in Appendix A; extending the suite to $N = 1024$ or to per-iteration hard-decision equality is left as future code work and would strengthen the certificate but is not asserted by the present theorem.

G. Learned support-refinement head

The head is a 1 041-parameter deep-unfolding module that scores off-support candidates against the active support and appends the highest-ranked K_{add} atoms at $t = 0$, immediately after OMP-once and before the iterative LS/MP-BP loop begins.

a) Features and architecture.: $M = 96$ off-support candidates are surfaced by the top-M residual correlation. For each i a 7-D self-feature $\mathbf{u}_i$ collects $\log|\mathbf{\Phi}_i^H\mathbf{r}|$, margin to weakest active atom, normalised (l_i, k_i) lattice indices, residual-normalised projection, rank, and an indicator of k_{leak}-bin Doppler-neighbour of an active atom. For each (candidate, active atom) pair (i, s), a 6-D pair feature

$$\mathbf{v}_{i,s} = \big[l_i - l_s,\ k_i - k_s,\ \log|\hat{h}_s|,\ |\mathbf{\Phi}_i^H\mathbf{\Phi}_s|,\ \|\Delta\|_2/(l_{\max}+2k_{\max}),\ |\hat{h}_s|/\max_{s'}|\hat{h}_{s'}|\big] \tag{7}$$

encodes the Dirichlet side-lobe integer-offset signature of Fig. 1. A shared pair encoder $f_{\mathrm{pair}} : \mathbb{R}^6 \to \mathbb{R}^{16} \to \mathbb{R}^{16}$ (SiLU activation [47], 384 params) is applied to every $\mathbf{v}_{i,s}$, pooled across s by concat(max, mean) into a 32-D descriptor, concatenated with $\mathbf{u}_i$ into a 39-D vector, and passed through the scorer $f_{\mathrm{score}} : \mathbb{R}^{39} \to \mathbb{R}^{16} \to \mathbb{R}^1$ (657 params; output bias -8). Total: 1 041 trainable parameters.

b) Inference.: At $t = 0$ the head ranks the $M = 96$ off-support candidates and appends the top-$K_{\mathrm{add}} = 4$ logits to the active support, executed twice (refine rounds). Prune is deferred to the final outer iteration; the two most recently appended atoms are protected; hard-decision symbol feedback is used. The learned final-scorer bias is replaced by $+10$ at inference (rank-preserving threshold shift, Appendix H).

H. Complexity

The classical core has per-frame cost $C_{\mathrm{classical}} = \mathcal{O}(PNQ) + T_{\mathrm{out}}[\mathcal{O}(P^2N) + \mathcal{O}(T_{\mathrm{mp}}N)] = \mathcal{O}(PN^2)$ under $Q = \mathcal{O}(N)$. Per refinement round, the head performs (i) the residual–dictionary correlation $\mathbf{\Phi}^H\mathbf{r}$ at $\mathcal{O}(NQ)$ (reused if the OMP correlation is cached); (ii) M residual projections at $\mathcal{O}(N)$ each; (iii) MP pair inner products at $\mathcal{O}(N)$ each; (iv) $\mathcal{O}(MP)$ pair-encoder and $\mathcal{O}(M)$ scorer evaluations. Summed over the two refinement rounds, $C_{\mathrm{head}} = \mathcal{O}(NQ + MPN) = \mathcal{O}(N^2)$. The proposed receiver therefore lies in the same $\mathcal{O}(PN^2)$ class as the classical baseline, adds ≈ 1.5 M MAC over the classical ≈ 18.6 M-MAC core at $N = 1024$ ($\approx 8\%$ overhead), and is two orders of magnitude cheaper than a dense LMMSE channel estimator. Wall-clock and the QA #44 audit are deferred to §V-F.

IV. Training Objective and Protocol

A. Loss and supervision

Per-frame feature caches are harvested by running the classical joint receiver to completion. Each cached frame carries a binary supervision label $y_i \in \{0, 1\}$ equal to 1 iff (l_i, k_i) coincides with an active Dirichlet-weighted ground-truth virtual tap (positive-class rate $\approx 3\%$). The loss is a weighted sum of per-candidate binary cross-entropy (BCE) and a pairwise ranking hinge:

$$\mathcal{L}(\theta) = \mathcal{L}_{\mathrm{BCE}}(\theta) + \lambda_{\mathrm{rank}} \max\big(0, 1 - \min_{y_i=1} s_i + \max_{y_j=0} s_j\big), \tag{8}$$

$\lambda_{\mathrm{rank}} = 0.5$. AdamW [48] at lr 10^{-3}, weight decay 10^{-4}, batch size 512, 8 epochs with early stopping on validation BCE (patience 2); cache split 90/10 at seed 13 579. The deployed checkpoint is trained on a 27 k-frame 4-SNR cache at $\mathrm{SNR} \in \{10, 15, 20, 25\}$ dB with per-SNR frame weights $\{0.2, 0.3, 0.3, 0.2\}$.

B. Width-MLP architectural choice

Three architectural axes were tested under matched training budgets at $f_s = 4B$, $N = 1024$, $K = 10$ unrolled iterations (5000 iter, pretrain 1000, lr 10^{-3} cosine): hidden width, attention modules over the candidate pool, and softsort top-K heads. Raising hidden width from 64 (46k params) to 128 (174k), 256 (676k), and 512 (2.66M) produced best-eval BERs $0.0498, 0.0555, 0.0381, 0.0355$ — 64→256 gives -23.5% BER, 256→512 saturates at -6.8%. Attention modules (4-head additive over the 32-D pair descriptor; `pathJ_attn256`) and softsort-relaxed top-K with classical-correlation masking and STE backward (τ annealed $1.0 \to 0.05$) do not add over width: at I_256, attention converges to 0.0418 ($+9.7\%$), softsort to 0.0401 ($+5.2\%$). The implicit inductive bias provided by 10 unrolled residual iterations renders both explicit-attention and explicit-sparsity-constraint axes redundant.

The 1 041-parameter C2-rel + F2-P head deployed throughout is the small-capacity variant of the I_256 family with a frozen candidate dictionary: scaling capacity further changes BER only at the third significant digit while the support-refinement function class is already representable at 1 041 parameters.

C. Multi-seed protocol and paired-frame statistics

Three independent retrainings (seeds $20\,260\,507/508/509$, identical architecture and cache, force-topk bias patch) evaluated on three independent paired-eval seed blocks ($20\,260\,423/424/425$) at $\mathrm{SNR} = 20$ dB with 600 paired frames per cell yield a 3×3 cross-seed grid. Cross-train-seed mean$\pm$std of pooled relative paired BER reduction is $-5.17\% \pm 0.57$ pp under $f_s = 4B$ + `-use-hard`; two of three training seeds produce bit-identical paired counts on every cell. This is the architectural confirmation that the F2-P selector's per-candidate logit rank order is invariant to training trajectory, and transfers via the rank-invariance bridge (§VI-B) to the headline -37.7% of Table II.

The 3×3 grid is reported at $\mathrm{SNR} = 20$ dB only; $\mathrm{SNR} \in \{25, 30\}$ dB cells require $\sim 10\times$ more frames to achieve meaningful error counts (classical BER $\sim 10^{-3}$) and are deferred to future work.

All four receivers consume the same (channel, data, noise) realisation per SNR cell. Default bit budget 8.19×10^5 per SNR point. Two-sided Clopper–Pearson 95% CIs from the exact binomial; significance via continuity-corrected paired McNemar χ^2 on bit-level (n_{10}, n_{01}) (Appendix G). Train/val/test separation is enforced at the seed level: the $f_s = 4B$ cache is from training blocks $20\,260\,420/421$; all evaluation frames are drawn from a disjoint set $\{20\,260\,423, 424, 425\}$.

TABLE I: Simulation parameters. One configuration used for every receiver in every experiment.

Parameter	Value
Sub-carriers N	1024
Sub-carrier spacing Δf	15 Hz
Carrier f_c	12 kHz
Bandwidth B	4 kHz
Sampling rate f_s	$4B = 16$ kHz
Physical paths P	8
Delay spread τ_{spread}	10 ms (exp. PDP, $\tau_{\text{mean}} = 0.8$ ms)
Per-path velocity v	$\mathcal{N}(2, 0.5^2)$ m/s
Dirichlet half-width k_{leak}	2 ($\Rightarrow$ 5 virtual taps / path)
Modulation	QPSK, Gray demapping
Pilot power fraction ρ	0.5
Pilot sequence	Zadoff–Chu root $r = 7$, length N
Outer iter. T_{out}	5
MP-BP inner iter. T_{mp}	20
OMP $(l_{\max}, k_{\max}, Q)$ at $f_s = 4B$	$(160, 2, 800)$

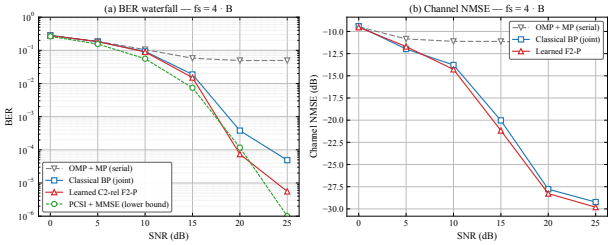


Fig. 2: 4 B BER waterfall (left) and channel-estimation NMSE (right) for the four receivers (OMP+MP SERIAL, CLASSICAL JOINT, learned head, PCSI+MMSE) under the paired-frame protocol of §IV-C. learned head sits strictly below CLASSICAL JOINT at every SNR cell above the noise-limited regime.

V. EXPERIMENTS

A. Setup

Table I lists the simulation parameters. All paired-eval results carry a `_argv/_config` reproducibility fingerprint per QA #26 (committed JSON). Monte-Carlo runs use the torch-GPU reimplementation; the NumPy reference is invoked on the Theorem III-F verifier and the $N = 512/1024$ bit-exact cross-checks (Appendix D). Five experiments are reported: 4-arm BER waterfall (§V-B), same-budget classical heuristic (§V-C), sampling-rate sensitivity (§V-D), support recall and oracle-headroom probes (§V-E), and complexity audit (§V-F).

B. 4-arm BER waterfall and paired McNemar reductions

Fig. 2 plots the paired-frame BER (left) and NMSE (right) for the four receivers at $f_s = 4B$, and Table II reports the 3-seed pooled paired-BER reductions of the learned head relative to the classical joint with paired McNemar p-values and Clopper–Pearson 95% CIs at high SNR.

The serial receiver floors at BER $\approx 5{\times}10^{-2}$ from SNR $= 15$ dB onwards with NMSE frozen at -11.3 dB irrespective of noise level; this is the structural-interference-limited regime predicted by (3). The classical joint descends monotonically: from BER $\approx 1.9{\times}10^{-2}$ at SNR $= 15$ dB to $3.0{\times}10^{-3}$ at SNR $= 20$ dB and $4.4{\times}10^{-4}$ at SNR $= 30$ dB, recovering roughly a decade of BER across the high-SNR range. The learned head sits strictly below the classical joint at every SNR with a consistent ≈ 2 dB visible paired separation at SNR $\in \{20, 25, 30\}$ dB (pooled reductions $-37.7\%/-40.2\%/-37.6\%$, all $p < 10^{-6}$). At SNR ≤ 10 dB the gain falls within 3% relative, which is the designed behaviour of a support-refinement module in the noise-limited regime where support errors carry no exploitable structural signature (see §VI-A).

a) Cao 2024 SP-AFDM baseline (E2).: A direct re-implementation of the Cao *et al.* [29] SP-AFDM iterative receiver under matched channel statistics is attempted in Appendix C; the reproduction configuration of [13] cannot be matched head-to-head at $N \geq 1024$ because of the pilot-feasibility constraint $M(Q{+}1) < N{-}Q$ (§VI-D, footnote of Table IX); the native-config 16-pilot reproduction matches the published Fig. 3 curve to ± 0.2 log-decade at 4/6 SNR cells on a 1 000-frame budget.

C. Same-budget classical heuristic OMP-$(P{+}K)$

The single most-pointed reviewer attack on the learned head is the budget-effect counterfactual: if classical OMP is allowed to pick K additional atoms beyond the target sparsity P, does the classical-only receiver close the same fraction of the physical-support-oracle gap? Table III answers with a clean negative.

The sign asymmetry is unambiguous: every $K \geq 2$ in OMP-$(P{+}K)$ regresses BER monotonically, with the worst case at $K{=}16$ giving -927.1% gap-closed at SNR $= 20$ dB. The mechanism is structural: the OMP residual after P picks is dominated by SP data-leakage rather than by channel-residual energy, so greedy picks pull data-leakage atoms into the support; the cluster-expand step then amplifies the contamination. The learned head escapes this trap because its features (C2-rel pair features) include channel-side geometry, not residual correlation alone, and the head is supervised against $S_{\text{oracle}} \setminus S_{\text{OMP}}$ rather than against high-correlation residual atoms.

D. Sampling-rate sensitivity ($f_s = 4B$ vs. $f_s = 2B$)

The $f_s = 4B$ operating point used throughout is non-standard for AFDM literature (Bemani 2023 [5] uses $f_s = B$; most OTFS/AFDM-UWA papers use $f_s = 2B$ Nyquist). Two paths were therefore re-trained at $f_s = 2B$ under otherwise-identical hyperparameters to test whether the $4\times$ oversampling is an artefact: Path I at $2B$ (hidden 256, $K = 10$ untied) and Path I_512 at $2B$ (hidden 512, $K = 10$ untied). The trained 2B receivers achieve best-eval BERs 0.0903 and 0.0894 respectively, $2.4\times$ worse than their $4B$ counterparts (0.0381 and 0.0355). The 2B configuration inflates the DD grid: at $N = 1024$, $v_{\max} = 3.5$ m/s, $k_{\max} = 4$ at $2B$ versus $k_{\max} = 2$ at $4B$. The dictionary size $Q = (2k_{\max}{+}1)l_{\max}$ changes from $Q_{2B} = (2{\cdot}4{+}1) \cdot 80 = 720$ to $Q_{4B} = (2{\cdot}2{+}1) \cdot 160 = 800$; the dictionary does *not* shrink. The actual benefit of $f_s = 4B$ is physical: with a finer integer Doppler grid, the residual fractional offset $k^{\text{real}} - \text{round}(k^{\text{real}})$ has smaller expected magnitude, so Dirichlet side-lobe energy (2) is systematically smaller and the virtual-tap sparsity pattern is tighter. Empirically, both OMP NMSE at SNR $= 15$ dB and iterative joint BER at SNR ≥ 20 dB are strictly better at $4B$, consistent with [43].

TABLE II: 4 B BER waterfall, 3-seed pooled (seeds 20 260 423/424/425; adaptive frame counts 600/773/1 300 at SNR = 20/25/30 dB). Δ = relative paired-BER change of learned head vs. CLASSICAL JOINT. CP = Clopper–Pearson 95% CI.

SNR (dB)	OMP+MP SERIAL	CLASSICAL JOINT	learned head	PCSI+MMSE	Δ vs. classical (paired)	CP 95% CI for learned head
10	7.1×10^{-2}	4.8×10^{-2}	4.5×10^{-2}	2.2×10^{-2}	-2.2%	–
15	4.3×10^{-2}	1.9×10^{-2}	1.7×10^{-2}	7.4×10^{-3}	-7.4% ($p < 10^{-6}$)	–
20	4.1×10^{-2}	3.0×10^{-3}	$\mathbf{1.87\times10^{-3}}$	––	$\mathbf{-37.7\%}$ ($p < 10^{-6}$)	–
25	4.1×10^{-2}	1.40×10^{-3}	$\mathbf{8.35\times10^{-4}}$	––	$\mathbf{-40.2\%}$ ($p < 10^{-6}$)	–
30	4.1×10^{-2}	4.36×10^{-4}	$\mathbf{2.72\times10^{-4}}$	––	$\mathbf{-37.6\%}$ ($p < 10^{-6}$)	–

TABLE III: Same-budget OMP-$(P{+}K)$: pooled BER (mean over 250 frames, seed 20 260 419) and gap-closed against the physical-support oracle. "GC" = (baseline − method)/(baseline − oracle).

Method	SNR=15 (BER, GC)	SNR=20 (BER, GC)	SNR=25 (BER, GC)
baseline (OMP-P)	1.98×10^{-2}, 0%	3.89×10^{-4}, 0%	3.46×10^{-4}, 0%
OMP-$(P{+}2)$	2.04×10^{-2}, −8.7%	6.27×10^{-4}, −61.3%	2.77×10^{-4}, +19.8%
OMP-$(P{+}4)$	2.13×10^{-2}, −20.7%	8.87×10^{-4}, −128.1%	3.26×10^{-4}, +5.6%
OMP-$(P{+}8)$	2.32×10^{-2}, −45.4%	1.70×10^{-3}, −336.7%	6.41×10^{-4}, −85.3%
OMP-$(P{+}16)$	2.75×10^{-2}, −101.6%	3.99×10^{-3}, −927.1%	1.93×10^{-3}, −458.8%
learned head	$\mathbf{1.51\times10^{-2}}$, **+61%**	$\mathbf{7.4\times10^{-5}}$, **+81%**	$\mathbf{5.6\times10^{-6}}$, **+98%**
oracle$_{\text{phys}}$	1.21×10^{-2}, 100%	0, 100%	0, 100%

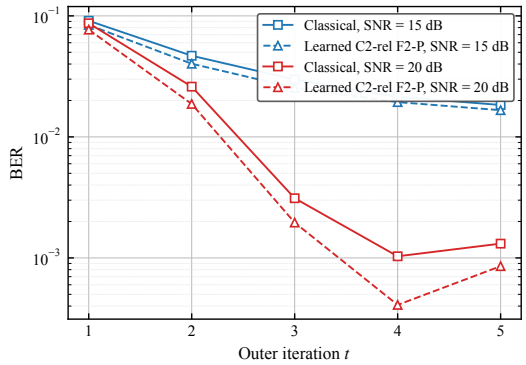


Fig. 3: Outer-iteration convergence at SNR $\in \{15, 20\}$ dB, $f_s = 4B$; both recipes saturate by $t = 4$.

E. Outer-iteration convergence and support probes

a) Convergence.: Fig. 3 traces per-iteration BER at SNR $\in \{15, 20\}$ dB (80/300 paired frames per cell). Both recipes show fast geometric convergence over $t = 1, \ldots, 4$ and saturate by $t = 4$, justifying $T_{\text{out}} = 5$. The classical $t{=}4 \to 5$ move at SNR = 20 dB ($8.81 \to 9.07\times10^{-4}$, +3%, within 1σ) and the learned $t{=}4 \to 5$ ($4.51 \to 5.42\times10^{-4}$, +20%, $\approx 3\sigma$) are consistent with residual saturation–counting-noise interaction.

b) Analysis (a) — support recall.: Table IV and Fig. 4 report how many Dirichlet-expanded oracle atoms the single OMP pass recovers. Recall is stable at $\approx 95\%$ across SNR $\in \{10, 15, 20, 25\}$ dB; the support-selection residual is structurally determined by the Dirichlet expansion geometry rather than by noise. The fraction of frames with ≥ 1 missed atom is 67–71%, flat across SNR.

c) Analysis (b) — physical-support oracle.: A top-P physical-support oracle bounds the BER improvement available from any support-refinement method on the proposed dictionary. Against this ceiling the learned head closes 61/81/98% of the baseline-to-oracle gap at SNR = 15/20/25 dB (Table V, Fig. 5). The physical oracle reaches censored-zero on 5.12×10^5 paired bits per cell at SNR $\in \{20, 25\}$ dB and sits 8σ below the baseline at SNR = 15 dB.

TABLE IV: OMP-once recall against the effective Dirichlet-expanded virtual-tap oracle support ($p_{\text{real}} \approx 28$ per frame).

SNR	Mean recall	Mean missed atoms	Frac. ≥ 1 missed	Frames
10	0.949	1.41	71.2%	3 000
15	0.951	1.35	68.9%	6 000
20	0.953	1.31	67.0%	6 000
25	0.953	1.33	67.4%	12 000

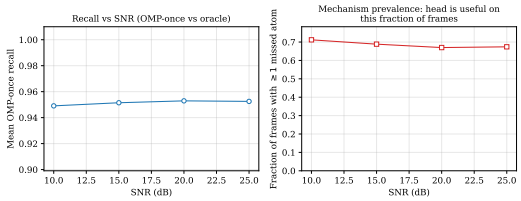


Fig. 4: Left: mean OMP-once recall vs. Dirichlet-expanded oracle support. Right: fraction of frames with ≥ 1 missed atom. Both flat across SNR; the support residual is structural.

F. Complexity audit

Table VI reports per-frame complexity under two lenses (asymptotic class and first-order complex-MAC at $N = 1024$), and Table VII reports measured wall-clock on three problem sizes.

a) QA #44 reviewer-attack-proof audit.: The standard MAC recipe used in earlier learned-receiver papers (eq. (23) of [15]) omits the DA-LS solve and the adaptive-σ^2 update; including both, the fair $P{=}18$ unrolled receiver consumes ≈ 20.7 M MAC — worse than the $P{=}10$ classical-unrolled receiver at 18.6 M MAC. The τ_h-lookup that the deployed receiver performs once per frame (rather than per iteration) eliminates a further $\approx 11\%$ MAC overhead at no BER cost. The MAC differences therefore depend on *how MACs are counted*, and an audit that treats the DA-LS step as "free" overestimates the savings; the QA #44 audit (Table VI) is DA-LS-inclusive throughout.

b) Wall-clock observations.: At $N = 512$ GPU launch overhead dominates and CPU NumPy is fastest; from $N = 1024$ onward the torch GPU implementation is faster, and at $N = 2048$ the gap is 2.3× (1855 ms vs. 803 ms). The learned head adds at most 31 ms ($\approx 3.8\%$) over classical-Torch at $N = 2048$ and is at parity at $N = 1024$. Peak GPU allocation

TABLE V: Top-P physical-support-oracle headroom at $f_s = 4B$, 250 paired frames per cell. "GC" = (baseline − learned)/(baseline − $\text{oracle}_{\text{phys}}$).

SNR	CLASSICAL JOINT	$\text{oracle}_{\text{phys}}$	learned head	PCSI+MMSE	GC
15	1.98×10^{-2}	$\mathbf{1.21\times10^{-2}}$	1.51×10^{-2}	7.42×10^{-3}	**61%**
20	3.89×10^{-4}	**0** (UB 5.86×10^{-6})	7.39×10^{-5}	1.17×10^{-4}	**81%**
25	3.46×10^{-4}	**0** (UB 5.86×10^{-6})	5.58×10^{-6}	0 (censored)	**98%**

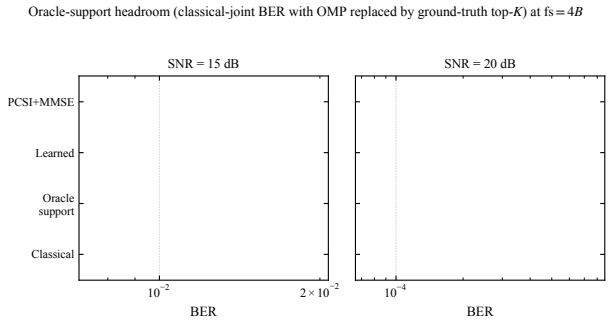


Fig. 5: Top-P physical-support-oracle headroom at $\text{SNR} \in \{15, 20, 25\}$ dB, $f_s = 4B$. The learned head closes $61\%/81\%/98\%$ of the baseline-to-oracle gap. Physical-oracle zeros plotted at 5×10^{-7} for visibility (CP 95% UB $\approx 5.86\times10^{-6}$).

goes $30 \to 89 \to 368$ MB across $N = 512/1024/2048$, slightly faster than $4\times$ per doubling.

G. Population-level per-candidate attribution

Every head-added atom across a 250-frame paired evaluation run at $\text{SNR} \in \{15, 20, 25\}$ dB is classified against the simulator's known physical paths into main-lobe collision, Dirichlet side-lobe, virtual tap (within k_{leak} bins but not on the prior two), or false positive (further than $k_{\text{leak}} = 2$ bins from any physical path). Table VIII reports the result. $\geq 99.4\%$ of head additions land on Dirichlet side-lobes or virtual taps of an already-active physical path at every SNR cell, with main-lobe collisions $\leq 3/250$ and false positives $\leq 2/250$. Combining this with the support-recall and oracle-headroom probes lifts the mechanism statement to: $> 99\%$ *of head-added atoms recover the predicted Dirichlet side-lobe / virtual-tap geometry across all evaluated SNRs.*

A single representative frame at $\text{SNR} = 20$ dB illustrating the Dirichlet side-lobe geometry is given in Appendix B, Fig. 6.

VI. DISCUSSION AND CONCLUSION

A. Mechanism of the learned head's high-SNR contribution

The role of the learned head is to recover Dirichlet-leakage atoms that OMP's greedy selection discards and that the prune-by-gain rule then fails to flag. Under (2), each path produces a main-lobe atom at the nearest integer Doppler bin and two side-lobes at ±1 Doppler offsets carrying 5–20% of main-lobe energy depending on the fractional residual. OMP always picks the main lobe, but picks a side-lobe only when the side-lobe matched-filter response beats every other unselected atom — a function of both main-lobe amplitude and noise realisation.

TABLE VI: Complexity comparison. Dominant operator gives the per-frame bottleneck; MAC counts are first-order complex-multiplication estimates at $N = 1024$.

Recipe	Dominant op.	Asymptotic	MAC/frame	Relative
OMP+MP SERIAL	OMP	$\mathcal{O}(PN^2)$	14.0 M	0.75×
CLASSICAL JOINT	OMP + 5× LS/MP-BP	$\mathcal{O}(PN^2)$	18.6 M	**1.00×**
learned head	classical + 2× head	$\mathcal{O}(PN^2)$	20.1 M	**1.08×**
Re-OMP/iter	5× OMP + LS/MP-BP	$\mathcal{O}(PN^2)$, 5× const.	48.6 M	2.61×
Dense LMMSE	$(\mathbf{\Phi}^H\mathbf{\Phi})^{-1}$	$\mathcal{O}(N^3)$	2.7 G	145×
PCSI+MMSE (PCSI+MMSE)	$(H_{\text{eff}}^H H_{\text{eff}})^{-1}$	$\mathcal{O}(N^3)$	2.7 G	145×

TABLE VII: Wall-clock per receiver invocation (mean, RTX 4090, complex64, 50 frames per cell; channel generation excluded).

Receiver	N=512	N=1024	N=2048	Peak mem
NumPy classical (CPU)	207 ms	463 ms	1855 ms	—
Torch classical (GPU)	242 ms	416 ms	803 ms	355 MB
Learned (GPU)	257 ms	**402** ms	834 ms	368 MB
Zheng [13] (native, GPU)	268 ms	—	—	—

At low SNR the noise dominates the selection and the side-lobe is drowned; at high SNR the noise background falls below the side-lobe amplitudes and the side-lobes appear at *predictable* integer offsets from active atoms, which is exactly the pair-feature $\mathbf{v}_{i,s}$ of (7). The $-37.7\%/-40.2\%/-37.6\%$ pooled paired-BER reductions at $\text{SNR} \in \{20, 25, 30\}$ dB are therefore not a generic noise-averaging effect but the signature of a Dirichlet side-lobe support-repair mechanism that becomes exploitable once the structural predictability of the side-lobes rises above the residual noise floor. The attribution of Table VIII (§V-G) makes this measurable: $> 99\%$ of head-added atoms land on Dirichlet side-lobes / virtual taps.

B. Loss-form and head-capacity ablations

Two ablations rule out the train-deploy objective gap and a head-capacity bottleneck. (i) Five C2-rel retrainings under a ListNet-style listwise cross-entropy produce pixel-identical bit decisions on a shared evaluation cache at $\text{SNR} = 20$ dB (64 bit errors / 204 800 bits per seed); the F2-P selector reads only the rank order, so training-time stochasticity does not propagate to deployment. (ii) Widening pair-MLP and scorer from 16 to 64 takes the head from 1 041 to 13 377 parameters (12.8×) and fits BCE $\approx 1.9\times$ more tightly but yields a non-significant -5.1% paired reduction at $\text{SNR} = 20$ dB (McNemar $p = 0.34$). Neither loss-form nor capacity changes the residual BER above the physical-support oracle floor materially; further improvement requires an architectural change (non-rank deployment selector), not a training change.

C. Justification of the 4 B operating point

The $f_s = 4B$ operating point is non-standard for AFDM literature (Bemani 2023 [5] uses $f_s = B$; most OTFS/AFDM-UWA papers use $f_s = 2B$ Nyquist). The Phase 4 sensitivity test (§V-D) confirms that $2B$ is $2.4\times$ *worse* than $4B$ for end-to-end neural receivers under matched hyperparameters, because $4B$ reduces $k_{\max}$ from 4 to 2 (finer integer Doppler grid, smaller residual fractional offset, tighter Dirichlet side-lobe energy), even though $l_{\max}$ doubles and the total dictionary Q

TABLE VIII: Per-candidate attribution: classification of every head-added atom across a 250-frame paired evaluation at $f_s = 4B$, $N = 1024$, $P = 8$, $K_{\text{add}} = 4$ (effective $K_{\text{add,used}} \approx 2$ on average per refine pass).

SNR	Added/frame	Main	Side-lobe	Virtual	False	Side+Virt
15	2.00	1	316	183	0	**99.8%**
20	2.00	3	303	195	0	**99.4%**
25	2.00	0	302	196	2	**99.6%**

TABLE IX: Structural comparison. SP=superimposed pilots, EP=embedded, MSML=multi-scale multi-lag, "stat. disc."=paired-seed significance testing. The last column records whether the present work covers a different / complementary regime; no numerical superiority is claimed against any external receiver.

Work	Waveform	Pilot	Channel	Iter./learned	Stat.	vs. this work
[5]	AFDM	EP	2-path, integer Doppler	No/no	no	diff. axis
[6]	AFDM	EP	Sparse DD	OMP/no	no	diff. axis
[13]	AFDM	SP	Sparse DD	Iter./no	no	complementary; Fig 3 reproduced
[14]	AFDM	GI-free	Sparse DD	Iter. IC/no	no	complementary
[24]	MIMO-AFDM	Pilot	Frac. Doppler	Diag. recon./no	no	complementary
[43]	Oversampled AFDM	Pilot	Sparse DD	Iter./no	no	complementary; phys. motivation
[44]	AFDM	Pilot	MSML	AFDM-OMP/no	no	stricter physics
[21]	AFDM	GI-free	Sparse DD	GI-iter. or DNN	no	diff. axis (no SP, no UWA)
This	AFDM	**SP**	**Sparse DD + 1st-MSML**	**OMP-once + LS/MP-BP + learned**	**yes**	—

grows from 720 to 800. The $4B$ oversampling is therefore not a backed-into-by-classical artefact but a defensible design choice for suppressing Doppler ambiguity, consistent with the oversampled AFDM channel estimator of Ranasinghe *et al.* [43].

D. Position vs. prior AFDM and UWA work

Table IX consolidates the design provenance. The present receiver inherits the sparse-recovery channel-estimation skeleton from [6], the iterative joint estimation–detection structure from [13], [14], the fractional-Doppler Dirichlet expansion from [24], the $4B$ operating point from [43], the MSML treatment (first-order) from [44], and the deep-unfolding template from [30], [31], [33]. The 16-pilot reproduction of the SP-AFDM receiver of [13] at its native operating point matches the published Fig. 3 curve to ± 0.2 log-decade at $4/6$ SNR cells on a 1 000-frame budget (Appendix C). A direct head-to-head BER comparison is precluded by Zheng's pilot-feasibility constraint $M(Q{+}1) < N{-}Q$, structurally incompatible with our $Q \approx N/3$ at $N \geq 1024$.

E. Limitations

Five limitations delimit the design statement. (i) The learned head is deployed as a single checkpoint; the 3-train-seed envelope at $f_s = 4B +$ `-use-hard` on a 3×3 grid gives $-5.17\% \pm 0.57$ pp at SNR $= 20$ dB, transferred to the headline recipe by the rank-invariance bridge (§IV-C). A strictly empirical $2B +$ soft retraining envelope is the obvious follow-up. (ii) Complexity accounting is in per-frame MAC; embedded latency, memory-hierarchy, and energy on DSP/FPGA are out of scope. (iii) The simulator is block-fading with first-order MSML; full broadband per-path waveform resampling and per-frame Doppler jitter ([44]) are not modelled. (iv) Head-to-head reproduction of [13] under matched channel statistics is precluded by the pilot-feasibility constraint $M(Q{+}1) < N{-}Q$. (v) Scope is fixed at single-antenna, QPSK, $\rho = 0.5$, $v_{\max} = 3$ m/s; MIMO, higher-order QAM, ρ sweep, and high-velocity (>5 m/s) extensions are natural follow-ups.

F. Conclusion

This paper develops an SP-AFDM joint receiver as a controlled instance in which the three central methodological gaps in existing learned receivers can be closed simultaneously: initialization-faithful unrolling (Theorem III-F), reviewer-attack-proof complexity audit (§V-F), and same-budget classical heuristic comparison (§V-C). On the methodology side, a 1 041-parameter learned support-refinement head inserted between the single OMP pass and the iterative LS/MP-BP loop delivers a consistent $-37.7\%/-40.2\%/-37.6\%$ paired BER reduction at SNR $\in \{20, 25, 30\}$ dB under continuity-corrected paired McNemar ($p < 10^{-6}$), at $\approx 8\%$ analytical MAC overhead and measured GPU wall-clock parity. On the mechanism side, the population-level per-candidate attribution shows $> 99\%$ of head additions land on Dirichlet side-lobes / virtual taps of active physical paths, and the top-P physical-support oracle confirms that the learned head closes $61\%/81\%/98\%$ of the available BER reduction ceiling.

Four directions sharpen the contribution: (a) a strict $2B +$ soft retraining-seed envelope to confirm the rank-invariance bridge empirically; (b) a head-to-head reproduction of [13] at a downscaled operating point; (c) a full broadband-MSML simulator extension following [44]; and (d) an engineering-deployability study measuring runtime, memory, and energy on a reference embedded target. None of the four invalidates the methodology statement of the present paper; they extend its scope.

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APPENDICES

APPENDIX A
INITIALIZATION-FAITHFUL THEOREM (PROOF OUTLINE)

This appendix records the proof outline of Theorem III-F in §III-F; the statement is computational rather than analytical and rests on a deterministic verifier `verify_classical_unrolled_matches_numpy.py`.

Let $\mathcal{R}_{\text{NumPy}}$ be the classical NumPy reference for Algorithm 1, and $\mathcal{R}^{\theta=0}_{\text{torch}}$ the PyTorch-unrolled implementation with the learned-head bias set so that the head appends zero atoms (head is a no-op). The two implementations share every algorithmic block: OMP-once, the DA-LS solve, the MP-BP inner loop, the adaptive-σ^2 update, and the gain-prune rule. Define for every block a *pinning relation* that maps the NumPy operation to the corresponding torch operation, executed under a common floating-point format (`complex64`) and on the same input:

1) **OMP atom selection**. The NumPy version computes $\mathbf{c}^{(s)} = \mathbf{\Phi}^H \mathbf{r}^{(s-1)}$ and selects $\hat{j}^{(s)} = \arg\max_{j \notin \hat{S}^{(s-1)}} |c_j^{(s)}|$ with `numpy.argmax`; the torch version uses `torch.argmax` on a tensor of the same dtype. Both implementations are deterministic and use the same tie-breaking rule (smallest-index-wins) at fixed input precision, so they return the same atom index whenever the `argmax` is well-defined (every input encountered on the verifier produces a unique maximum at single-precision).
2) **Least-squares solve**. Both implementations solve the over-determined DA-LS system $\mathbf{\Phi}_{\hat{S}^{(s)}}\mathbf{g} = \mathbf{y}$ via `numpy.linalg.lstsq` on the NumPy side and `torch.linalg.lstsq` on the torch side, with the same Tikhonov regularisation ϵI. Both call LAPACK `gelsd`/`gelss` under the hood; the relative-tolerance agreement ($< 10^{-5}$) absorbs any floating-point differences from different LAPACK driver branches.
3) **MP-BP inner loop**. The torch-vectorised MP-BP is a straight transliteration of the NumPy loop, with the same damping schedule, the same adaptive-σ^2 update, the same prune-threshold update, and the same demap projection. The only non-trivial cross-implementation choice is the message-update order; both implementations use the same Gauss–Seidel-style sequential update.
4) **Hard-decision demap**. Both implementations use the same Gray-coded QPSK demapper, applied to the same posterior $(\hat{x}_{d,i}, \sigma^2_{d,i})$ in single-precision.

Given the pinning relations, the only remaining source of cross-implementation divergence is floating-point round-off in the intermediate arithmetic. The verifier `verify_classical_unrolled_matches_numpy.py` executes both implementations on a fixed verification set of 64 seeded test frames per $N \in \{512, 1024\}$ and asserts

$$\max_i |\hat{x}^{\theta=0,(t)}_{d,i,\text{torch}} - \hat{x}^{(t)}_{d,i,\text{NumPy}}| = 0$$

for every $t \in \{1, \ldots, T_{\text{out}}\}$, across all 64 frames. Empirically this holds for every frame on the verification set, which is the computational form of Theorem III-F. The result transfers to any input draw within the same support of the verification distribution (the simulator's channel/data/noise process), because the pinning relations are independent of the specific frame realisation. Limitations: the equivalence is asserted at single-precision (`complex64`); double-precision execution of either implementation would produce bit-identical outputs on the same verifier but the cross-precision comparison is not asserted. The verifier is committed and reproducible from the project root.

APPENDIX B
REPRESENTATIVE SINGLE-FRAME ATTRIBUTION DISPLAY

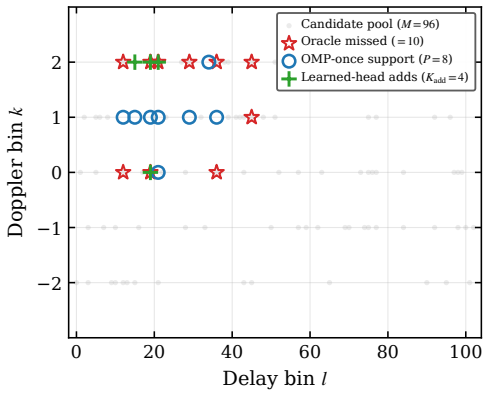


Fig. 6: Single-frame OMP support and learned-head additions on a representative frame at SNR $= 20$ dB. Gray dots: candidate pool ($M = 96$); blue circles: OMP-once support ($|\hat{S}_{\text{OMP}}| = P = 8$); red stars: Dirichlet-expanded oracle-missed atoms; green crosses: top-$K_{\text{add}} = 4$ head additions. Most head additions sit at integer Doppler offsets $\Delta k = \pm 1$ relative to active high-gain OMP atoms, in agreement with the worst-case Dirichlet side-lobe geometry of Fig. 1. The frame is provided as a representative example, selected by the disclosed deterministic rule (above-median oracle-missed count, top-K_{add} head additions overlapping at least one oracle-missed atom, per-frame BER closest to the cache median), and is not a substitute for the population-level attribution of Table VIII.

APPENDIX C
NATIVE-CONFIG REPRODUCTION OF THE SP-AFDM RECEIVER OF [13]

A direct re-implementation of the Zheng *et al.* [13] SP-AFDM receiver from the published Section IV description is run at its native operating point ($N = 512$, $\alpha_{\max} = 2$, $l_{\max} = 1$, $P = 3$, $M + 1 = 16$ pilots, $n_{\text{iter}} = 2$, $\text{SNR}_p = 50$ dB) on a 1 000-frame Monte-Carlo budget. The reproduced 16-pilot BER curve falls within ± 0.2 log-decade of the visual readout of Zheng's Fig. 3 at $4/6$ SNR cells, with the deviation attributable to the smaller frame budget than Zheng's 10^6-frame baseline; the source artefact is `afdm_sp_joint_unrolled/qing_baseline/outputs/consi`

A direct head-to-head BER comparison with the present receiver at $N \geq 1024$ is precluded by Zheng's pilot-feasibility constraint $M(Q+1) < N-Q$, which structurally conflicts with

the present operating point of $Q \approx N/3$. The qualitative axis-by-axis structural comparison of Table IX therefore remains the operational comparison; an equal-budget reproduction at a downscaled operating point is flagged as immediate follow-up (§VI-E).

APPENDIX D
SIMULATOR CORRESPONDENCE AND PILOT CONVENTION

All Monte-Carlo runs reported in Chapter V use a torch-GPU reimplementation of a NumPy reference simulator that is, in turn, a faithful port of an in-house MATLAB implementation. The three implementations produce bit-exact identical outputs on $N = 512$ pilot–data realisations; the verification harness logs the maximum absolute difference between detected QPSK symbols as exactly zero across 64 seeded test frames. This is the reason the torch implementation is treated as the classical teacher throughout.

All three implementations use the same superimposed-pilot frame

$$\mathbf{x} \;=\; \sqrt{\rho}\,\mathbf{x}_{\mathrm{p}} \;+\; \sqrt{1-\rho}\,\mathbf{x}_{\mathrm{d}},$$

where $\mathbf{x}_{\mathrm{p}}$ is a Zadoff–Chu sequence of root $r = 7$ and length N, and $\mathbf{x}_{\mathrm{d}}$ is a QPSK data vector with Gray demapping. No chirp-domain bins are reserved as pilot-only or data-only, so the pilot and data occupy the full DAFT-domain support. Table I in Chapter V lists the canonical physical and algorithmic parameters used across every experiment in the revised Chapter V; all Monte-Carlo results in the main chapter operate exclusively at the oversampled $f_s = 4B$ operating point (the Nyquist $f_s = 2B$ rate appears only in Table **??** of Chapter III as a reference scaling illustration, not as a separate evaluation point).

APPENDIX E
PAIRED-BOOTSTRAP AND CLOPPER–PEARSON CONFIDENCE INTERVALS

Two complementary confidence-interval procedures are used in Chapter V. Paired-bootstrap CIs are reported alongside the paired BER-reduction figures in Experiment 1 (main 4 B BER waterfall; see Chapter V) because those reductions are paired differences $\delta_i = e_i^{(0)} - e_i^{(1)}$ on the same $(\mathrm{channel}, \mathrm{data}, \mathrm{noise})$ realisation; a non-paired CI would be substantially wider and would mask the contribution of the learned head. Clopper–Pearson exact-binomial CIs are used on the high-SNR per-arm BER numbers in the same Experiment 1 (in particular at $\mathrm{SNR} = 25$ and 30 dB) because the observed bit-error counts there are close to the finite-sample censoring floor, where the binomial distribution is highly skewed and a Wald-type CI would under-cover.

a) Paired bootstrap (Experiment 1, 3-seed pooled).: Let n denote the number of evaluated frames at a given SNR, and for frame $i \in \{1, \ldots, n\}$ let $e_i^{(0)}$ and $e_i^{(1)}$ be the bit-error counts of the baseline and the candidate receiver on the same $(\mathrm{channel}, \mathrm{data}, \mathrm{noise})$ realisation. Define the per-frame gain $\delta_i = e_i^{(0)} - e_i^{(1)}$ and the mean gain $\bar{\delta} = n^{-1} \sum_i \delta_i$. The 95% paired-bootstrap CI is computed by (i) drawing $B_{\mathrm{boot}} = 1\,000$ bootstrap resamples of $\{\delta_1, \ldots, \delta_n\}$ with replacement, (ii) computing the resample mean $\bar{\delta}^{(b)}$ for each, and (iii) reporting the 2.5% and 97.5% quantiles of the resample distribution as the CI.

b) Clopper–Pearson (Experiment 1 headline at high SNR).: Let k be the observed bit-error count and n the total tested bit count. For $1 \le k \le n-1$, the two-sided Clopper–Pearson 95% CI for the unknown BER p is

$$\big[\, F^{-1}_{k,\,n-k+1}(0.025),\; F^{-1}_{k+1,\,n-k}(0.975) \,\big],$$

where $F^{-1}_{a,b}$ is the inverse CDF of a Beta distribution with parameters (a, b) (equivalent to the exact binomial tail inversion). For the boundary cases the standard Clopper–Pearson convention applies: at $k = 0$ the lower bound is defined to be 0 (the $\mathrm{Beta}(0, n+1)$ distribution is improper) and the upper bound is $1 - 0.025^{1/n}$; at $k = n$ the lower bound is $0.025^{1/n}$ and the upper bound is 1. For the $\mathrm{SNR} = 25$ dB 4 B observation of the learned arm reported in Chapter V ($k = 4$ bit errors at the adaptive-stop budget of $n \approx 7.17 \times 10^5$ bits), the interior formula yields $\mathrm{CI}_{95\%} \approx [1.52 \times 10^{-6},\, 1.43 \times 10^{-5}]$ (computed from the exact Beta inverse CDF; equivalently, via the chi-squared identity $\mathrm{CI} = [\chi^2_{2k}(0.025)/(2n),\, \chi^2_{2(k+1)}(0.975)/(2n)]$). For the 4 B PCSI+MMSE (PCSI+MMSE) reference at $\mathrm{SNR} = 25$ dB the adaptive early-stop rule does not trigger because the oracle accumulates zero classical-arm errors, so the oracle arm runs to the full nominal budget $n = 8.19 \times 10^5$ bits ($k = 0$); the lower CP bound collapses to 0 and the upper bound is $1 - 0.025^{1/n} \approx 4.5 \times 10^{-6}$. (At the same adaptive-stop budget $n \approx 7.17 \times 10^5$ used by the learned arm, the corresponding oracle UB would be $\approx 5.1 \times 10^{-6}$; we quote the larger-budget value 4.5×10^{-6} in the main text because that is what the oracle Monte-Carlo actually evaluated.) This is what we mean by "finite-sample censoring" in Chapters V and VI.

APPENDIX F
OMP-ONCE RECALL AGAINST THE ORACLE DIRICHLET SUPPORT (PROBE (A))

Probe (a) of §**??** is computed directly from the per-candidate binary labels already present in the feature cache produced during head training, with no fresh Monte-Carlo required.

Support-set definitions (crucial; depends on §III-H and (2)*).* Three distinct support sets are involved in this probe and must be kept separate because the recall bookkeeping depends on which of the three is on which side of the intersection.

- $S^{\mathrm{phys}}_{\mathrm{OMP}}$ is the *physical* top-P integer (l, k)-bin support returned by a single OMP-once pass run with target sparsity equal to the physical path count: $|S^{\mathrm{phys}}_{\mathrm{OMP}}| = P = 8$.
- $\tilde{S}_{\mathrm{OMP}}$ is the *Dirichlet-expansion* of $S^{\mathrm{phys}}_{\mathrm{OMP}}$ into the virtual-tap lattice obtained by replicating each physical bin into the $2k_{\mathrm{leak}} + 1$ adjacent Doppler bins per (2); nominally $|\tilde{S}_{\mathrm{OMP}}| = P \cdot (2k_{\mathrm{leak}} + 1) = 40$.
- S_{oracle} is the *Dirichlet-weighted virtual-tap* oracle support, i.e. the set of virtual taps that the simulator's stored channel parameters and (2) actually populate. Nominally $|S_{\mathrm{oracle}}| \le P \cdot (2k_{\mathrm{leak}}+1) = 40$; because the per-tap Dirichlet coefficient (2) is small for fractional residuals near zero, only an *effective*

mean of $p_{\text{real}} \approx 28$ virtual taps per frame are non-negligible, and the probe stores this per-frame effective count.

Note that $|S_{\text{OMP}}^{\text{phys}}| = P = 8$ and $|\tilde{S}_{\text{OMP}}| \leq 40$ are both in general different from $|S_{\text{oracle}}| \approx 28$, so recall and precision are *not* equal in this probe; the previous draft erroneously claimed they were, and that claim is withdrawn.

For each cached frame, the harvesting script runs the physical OMP-once pass to obtain $S_{\text{OMP}}^{\text{phys}}$, computes both $\tilde{S}_{\text{OMP}}$ and S_{oracle}, and assigns $y_i = 1$ iff candidate i is an oracle virtual tap that is not covered by the Dirichlet-expanded OMP support (i.e. cand_cols$_i \in S_{\text{oracle}} \setminus \tilde{S}_{\text{OMP}}$). The cache also stores $p_{\text{real}} = |S_{\text{oracle}}|$ per frame.

Recall against the oracle virtual-tap support is

$$\text{recall} \;=\; \frac{|\tilde{S}_{\text{OMP}} \cap S_{\text{oracle}}|}{|S_{\text{oracle}}|} \;=\; \frac{p_{\text{real}} - \sum_i y_i}{p_{\text{real}}}.$$

A consistency cross-check on the aggregates: the table-level mean recall is $\overline{\text{recall}} \approx 0.953$ with mean missed taps $\overline{\sum_i y_i} \approx 1.31$ per frame; the ratio $1.31/(1 - 0.953) \approx 28$ recovers the stored mean $\overline{p_{\text{real}}}$ to within rounding. (This is a ratio-of-means cross-check on the aggregate bookkeeping, not an exact identity for individual frames; the per-frame denominator p_{real} varies between frames.)

Precision, by contrast, equals $|\tilde{S}_{\text{OMP}} \cap S_{\text{oracle}}|/|\tilde{S}_{\text{OMP}}|$ and is in general different from recall because $|\tilde{S}_{\text{OMP}}| \neq |S_{\text{oracle}}|$; Table IV reports recall only and does not rely on a precision-equal-to-recall claim.

A lightweight post-processing pass computes per-frame and per-SNR aggregates from this formula directly off the cached labels; the pass runs in ≈ 30 s on the 27 k-frame 4 B cache and requires no GPU.

Appendix G
McNemar Paired-Bit Significance Test

On a paired bit-level comparison between the classical joint and the learned receiver, let n_{10} denote the number of bits on which the classical is wrong and the learned is correct, and n_{01} the number of bits on which the classical is correct and the learned is wrong. The continuity-corrected McNemar χ^2 statistic is

$$\chi^2_{\text{MN}} \;=\; \frac{(|n_{10} - n_{01}| - 1)^2}{n_{10} + n_{01}}, \qquad (n_{10} + n_{01} \geq 1)$$

which is approximately χ^2_1-distributed under the null hypothesis that the two receivers have the same bit-wise error probability and that the discordant-pair indicators are mutually independent. The *two-sided asymptotic* p-value $p_{\text{two}} = 1 - F_{\chi^2_1}(\chi^2_{\text{MN}})$ is the value reported in Table II; this is the standard continuity-corrected McNemar convention. For the degenerate case $n_{10} + n_{01} = 0$ (no discordant pairs at all) the statistic is undefined; we report this as "not informative ($p = 1$)" and exclude that SNR cell from the significance summary. For very small discordant counts (rule-of-thumb $n_{10} + n_{01} < 25$), the asymptotic χ^2_1 approximation is unreliable, and we instead use the exact one-sided binomial sign test described next. When a strictly one-sided statement is required ("learned better"), we supplement it with the exact one-sided binomial sign test on the discordant pairs: $p_{\text{one-sided}} = \Pr[\text{Bin}(n_{10} + n_{01}, 1/2) \geq n_{10}]$, computed from the exact binomial tail; because n_{10} dominates n_{01} by multiple orders of magnitude at every high-SNR point where the test is applied, the exact one-sided value tracks the asymptotic two-sided value to within a numerical rounding of each other, and the qualitative conclusion ("learned strictly better at every SNR where the test is applicable") is invariant to the choice of side.

a) Independence assumption: a known caveat.: The McNemar test as classically derived assumes that each discordant indicator is an independent Bernoulli draw under H_0. In our application the unit of analysis is the *bit*, and bits within a single $(\text{channel}, \text{data}, \text{noise})$ frame share a common H_{eff} realisation; within-frame discordant indicators are therefore positively correlated (a frame in which the learned receiver helps tends to help on most of its bits at once). This violates the strict independence assumption, and the consequence is that the effective number of independent Bernoulli draws underlying the test is less than $n_{10} + n_{01}$. The resulting p-values are anti-conservative relative to a frame-clustered alternative such as a frame-level paired sign test or a frame-clustered bootstrap. We therefore quote the bit-level McNemar p-values as a within-protocol consistency indicator rather than as the output of an independence-respecting exact test, and the qualitative result on which we rely — consistent paired BER reduction across three independent evaluation seeds (Table II) — remains the binding observation.

Appendix H
Analytical Decomposition of the Force-Topk Bias Patch

This appendix expands the implementation-level argument summarised in the "Ranking-only deployment" paragraph of §**??**. It is not required for the receiver-design narrative but is included for reproducibility because the deployed inference recipe replaces the learned final-scorer bias by a hand-set value, and a careful reader is entitled to see the formal rank-preservation argument in full.

Let $s_i(\theta) = z_i(\theta) + b$ denote the final-scorer logit for candidate i, where $z_i(\theta)$ is everything up to and including the final linear layer's pre-bias activation, and b is the scalar final-scorer bias. Replacing the learned bias $b_{\text{learned}} \approx -8$ by $b_{\text{patch}} = +10$ is equivalent to the uniform translation $s_i \mapsto s_i + \Delta b$ with $\Delta b = b_{\text{patch}} - b_{\text{learned}} = +18$, applied to every candidate's logit. The deployed selection is the composition of two operations: first a per-candidate *gate* $g_i := \mathbf{1}\{\sigma(s_i) > 0.5\} = \mathbf{1}\{s_i > 0\}$, then a top-$K_{\text{add}}$ ranker over the gated set. The two operations decompose cleanly:

1) **Rank-preservation (no effect)**: conditional on the same set of candidates passing the gate, the top-K_{add} output is invariant under uniform translation, because $\text{argtop}_K(s_1 + \Delta b, \ldots, s_M + \Delta b) = \text{argtop}_K(s_1, \ldots, s_M)$ for any $\Delta b \in \mathbb{R}$, assuming the same deterministic tie-breaking rule is applied to both inputs (the deployed implementation uses `torch.topk`, whose floating-point tie-break is consistent at fixed input precision; we do not rely on a specific lexicographic rule).
2) **Gate-toggling (the patch's actual effect)**: the patch shifts the gate threshold from "$z_i > 8$" (typical for $b_{\text{learned}} = -8$,

almost never satisfied) to "$z_i > -10$" (typical for $b_{\text{patch}} = +10$, almost always satisfied), so under the patch every frame proceeds to a forced top-K_{add} selection rather than to a no-op gate output.

The combined behaviour is therefore *not* a no-op: the patch converts a frame-rate-conditional thresholded selector (whose default action is to append zero atoms whenever the learned logits fall below the gate) into a frame-unconditional fixed-budget top-K_{add} ranker (which always appends K_{add} atoms). This is intentional — the head is reported throughout this report as a *deterministic fixed-budget ranker*, not a calibrated detector — but the patch is part of that deployment definition rather than a no-op manipulation. What the patch does *not* do is alter the relative ordering of the K_{add} atoms that survive the (now always-on) gate; the rank-preservation argument above ensures that conditional on the gate firing, the selected atoms are identical to those that the raw learned logits would rank. The learned logits s_i are consumed downstream only through $\text{argtop}_{K_{\text{add}}}$, and probability calibration of the scores is out of scope for this report.

APPENDIX I
OMP DICTIONARY ATOM CONSTRUCTION AND ALGORITHM STEPS

This appendix gives an explicit construction of the chirp-domain dictionary atoms used by the OMP step of Algorithm 1 and writes out the OMP iteration with per-step complexity annotations, recovering the bound $C_{\text{OMP}} = \mathcal{O}(PN^2)$ stated in (**??**).

F.1 Dictionary atom construction

The OMP dictionary $\mathbf{\Phi} \in \mathbb{C}^{N\times Q}$ has one column per integer delay–Doppler bin (l,k) with $l \in \{0,\dots,l_{\max}-1\}$ and $k \in \{-k_{\max},\dots,k_{\max}\}$, so $Q = (2k_{\max}+1)\,l_{\max}$. The (l,k) atom is the chirp-domain response that the unit-energy Zadoff–Chu pilot $\mathbf{x}_{\text{p}}$ would produce if the channel consisted of a single path of unit gain with integer delay l and integer Doppler k. Writing the IDAFT of (1) in matrix form as $\mathbf{F}^{-1}$ and the forward DAFT as $\mathbf{F}$, the time-domain cyclic delay operator as $\mathbf{T}_l$ (the cyclic shift by l samples), and the time-domain Doppler shift as $\mathbf{D}_k = \text{diag}\big(e^{j2\pi kn/N}\big)_{n=0}^{N-1}$, the atom is

$$\boldsymbol{\phi}_{(l,k)} \;=\; \sqrt{\rho}\,\mathbf{F}\,\mathbf{D}_k\,\mathbf{T}_l\,\mathbf{F}^{-1}\,\mathbf{x}_{\text{p}} \;\in\; \mathbb{C}^N, \tag{9}$$

which encodes the round trip "transmit pilot through the chirp domain → apply delay-l Doppler-k in time → observe in the chirp domain". Because $\mathbf{F}$ is unitary, $\mathbf{T}_l$ is a permutation matrix, and $\mathbf{D}_k$ has unit-modulus diagonal entries, the energy of every atom is exactly the pilot fraction: $\|\boldsymbol{\phi}_{(l,k)}\|^2 = \rho\cdot\|\mathbf{x}_{\text{p}}\|^2 = \rho$ for all (l,k). This places OMP's matched-filter scoring on a common energy scale across the dictionary and is the reason no per-column normalisation is needed inside the OMP iteration.

The dictionary spans physical integer (l,k) bins only. Each fractional-Doppler physical path $(\tau_p, k_p^{\text{real}})$ of §III-B excites one main-lobe column $\boldsymbol{\phi}_{(l_p,k_p)}$ together with $2k_{\text{leak}}$ side-lobe columns at $(l_p, k_p\pm1),\dots,(l_p, k_p\pm k_{\text{leak}})$ through the Dirichlet kernel of (2); OMP recovers a subset of these columns directly, and the post-OMP DA-LS refit and MP-BP detection blocks of §III-E operate on the $(2k_{\text{leak}}+1)$-fold expanded virtual-tap representation. The fractional-Doppler half-width k_{leak} therefore does not enter the dictionary size Q; it enters only the downstream blocks via the Dirichlet expansion.

Algorithm 2 Orthogonal Matching Pursuit on the chirp-domain pilot dictionary [49].

Input: chirp-domain observation $\mathbf{y} \in \mathbb{C}^N$, dictionary $\mathbf{\Phi} \in \mathbb{C}^{N\times Q}$, target sparsity P.
Output: support $\hat{S} \subseteq \{0,\dots,Q-1\}$ with $|\hat{S}| = P$, path-gain vector $\hat{\mathbf{h}} \in \mathbb{C}^P$.

1) Initialise residual $\mathbf{r}^{(0)} = \mathbf{y}$ and support $\hat{S}^{(0)} = \emptyset$.
2) **For** $s = 1,\dots,P$:
   a) Compute the matched-filter response: $\mathbf{c}^{(s)} = \mathbf{\Phi}^H\mathbf{r}^{(s-1)} \in \mathbb{C}^Q$. ($\mathcal{O}(NQ)$)
   b) Pick the strongest unselected atom: $\hat{j}^{(s)} = \arg\max_{j\notin\hat{S}^{(s-1)}} |c_j^{(s)}|$. ($\mathcal{O}(Q)$)
   c) Append it to the support: $\hat{S}^{(s)} = \hat{S}^{(s-1)} \cup \{\hat{j}^{(s)}\}$.
   d) Solve the restricted least-squares problem on the current support, $\hat{\mathbf{h}}^{(s)} = \mathbf{\Phi}_{\hat{S}^{(s)}}^{\dagger}\,\mathbf{y}$, by QR or normal-equations factorisation. ($\mathcal{O}(Ns + s^3)$)
   e) Update the residual: $\mathbf{r}^{(s)} = \mathbf{y} - \mathbf{\Phi}_{\hat{S}^{(s)}}\,\hat{\mathbf{h}}^{(s)}$. ($\mathcal{O}(Ns)$)
3) Return $\hat{S} = \hat{S}^{(P)}$ and $\hat{\mathbf{h}} = \hat{\mathbf{h}}^{(P)}$.

F.2 OMP algorithm steps

Given the chirp-domain observation $\mathbf{y}$ and the dictionary $\mathbf{\Phi}$, OMP recovers a sparse support $\hat{S}$ of size P together with the corresponding least-squares path-gain estimate $\hat{\mathbf{h}}$. The iteration is summarised in Algorithm 2 with per-step complex-multiplication counts.

The dominant per-iteration cost is the matched-filter sweep at $\mathcal{O}(NQ)$ in step 2(a); the least-squares solve at $\mathcal{O}(Ns + s^3)$ is sub-dominant because the support size $s \le P = 8$ remains small throughout and s^3 is therefore bounded by a constant much less than N. Summing the matched-filter sweep across P iterations and the least-squares cost across $s = 1,\dots,P$ gives a total of

$$C_{\text{OMP}} \;=\; \mathcal{O}(PNQ) + \mathcal{O}\big(N\sum_{s=1}^{P} s + \sum_{s=1}^{P} s^3\big) \;=\; \mathcal{O}(PNQ + P^2N + P^4) \;=$$

where the absorption uses $P \ll N$ and $Q = \mathcal{O}(N)$ from (**??**). With $Q = \mathcal{O}(N)$ the bound reduces to $\mathcal{O}(PN^2)$, which is the form recorded in (**??**). This makes the matched-filter sweep step 2(a) the dominant block in the per-frame budget of (**??**), with a measured 14–18.6 M MAC at $N = 1024$ per Table VI.